Cross-Lingual Prompting and Trace Interventions: Draft Sections for the Final Report

Team Netherite. Sections on RQ4 (cross-lingual prompting) and RQ5 (trace interventions), with all numbers generated from the analysis outputs in the repository. 31 October 2026.

Notes for editing (delete before submission)

Sources. Part A: experiments/2026-10-31_clp-analysis-rahul (outputs/numbers.json, outputs/analysis.md). Part B: experiments/2026-10-27_trace-deletion-selfhost (outputs/selfhost_G, selfhost_S, noise_and_loops.json) and experiments/2026-10-29_distractor-resampling (outputs/resample/analysis.json, sensitivity.json). Every table and every number in the text is filled from those files by the script that built this document; nothing was typed in by hand except counts that define the design (2,250 questions, 4,096 tokens, 7 rounds and similar).

What changed since the previous trace draft. Numbers marked [sensitivity] there now come from scripts in the repository (sensitivity.py, noise_and_loops.py). Three corrections: unanswered continuations in Experiment 1 are 102 (14.5%) under greedy and 40 (5.7%) under sampling when cut-off outputs count as unanswered, as in the experiment's analysis (the old draft gave 99 and 37, the count without an Answer line); cut-off continuations in Experiment 2 were 9.4–11.9% per round (old: about 9–11%); answered-only intervals moved by at most 0.2 points because the script uses the experiment's own bootstrap settings.

Context from the full-grid analysis (feat/full-grid-analysis, FINDINGS.md). On all 6,750 questions, the Hindi CoT distraction penalty for the time-related distractor is +2.2 [+0.6, +3.8]. On the 2,250 questions with CLP responses it is smaller and not significant (Section A.5). Part A only compares conditions on the same 2,250 questions.

Not included (done by teammates): edit-distance (RDS) analysis, the logistic regression and the en-CoT condition. **Citations marked [verify]** need their bibliographic entries checked. **[EDIT]** marks places to align with the rest of the report.

For the presentation. Appendix A lists every claim with its evidence and strength. Be ready to explain: what the two CLP stages do; why stage 1 matters (Table A3); the four resampling arms (Table B2); and the limitations of both parts.

Part A. RQ4: Does cross-lingual prompting help Hindi temporal reasoning?

A.1 Method

Cross-lingual prompting (CLP; Qin et al., 2023) solves a non-English question in two turns. In **stage 1** the model is asked to "act as an expert in multi-lingual understanding in Hindi" and to "understand the task in English step-by-step". In **stage 2**, in the same conversation, it is asked to "resolve the task you understand above step-by-step" with its reasoning entirely in English and a final Answer line. We ran CLP on the Hindi questions with the main study's settings (Sarvam-105B through the API, thinking off, temperature 0, up to 4,096 tokens per stage). Day-of-week answers are requested as Hindi weekday names and scored by weekday, as in the main study.

CLP was run on questions 0–49 of every task and difficulty level: 2,250 questions, each with no distractor, a time-related distractor and an unrelated distractor, so 6,750 responses. We compare it with four conditions from the main runs on exactly the same questions: English and Hindi direct answers and English and Hindi CoT.

We apply the mid-report analysis pipeline as rerun on the full grid: same answer extraction, scoring, error classes, distractor-mention rule and statistics (exact McNemar tests on paired responses with Holm correction over tasks; 95% confidence intervals from a 2,000-sample bootstrap over questions). On the 27,000 standard responses our per-response data is identical to the full-grid pipeline's, and an independent script that shares no code reproduces each condition's overall, per-distractor and selected per-task accuracies and the McNemar counts for CLP. The CLP records have the same question text and gold answer as the standard Hindi records in every case.

A.2 Accuracy

Task	English direct	English CoT	Hindi direct	Hindi CoT	Hindi CLP	CLP − Hindi CoT	CLP − Hindi direct
Date addition	85.6	98.0	82.8	85.1	92.7	+7.6*	+9.9*
Date subtraction	86.0	95.2	84.8	66.0	86.9	+20.9*	+2.1
Date duration	84.0	99.3	78.9	88.3	92.7	+4.4*	+13.7*
Date recurrence	22.1	90.3	2.8	20.1	27.0	+6.8*	+24.2*
Day of week	14.9	69.5	15.2	21.0	31.8	+10.8*	+16.6*
Interval date	43.2	50.7	46.8	23.9	35.3	+11.4*	−11.5*
Time addition	70.5	95.9	75.6	73.5	88.2	+14.7*	+12.6*
Time subtraction	64.7	97.7	64.9	75.2	90.1	+14.9*	+25.2*
Time duration	65.7	98.9	44.7	80.1	86.7	+6.6*	+42.0*
All tasks	59.7	88.9	55.2	59.4	70.3	+10.9*	+15.1*

Table A1. Accuracy (%) of answered responses on 2,250 questions × 3 distractor conditions, pooled over difficulty and distractor condition. Differences in points; * marks $p < 0.05$ in an exact McNemar test on paired responses, Holm-corrected over the nine tasks (the All row is uncorrected).

CLP helps Hindi substantially. CLP answers 70.3% of Hindi questions correctly, against 59.4% with Hindi CoT (+10.9 [+9.6, +12.2] points; 1,297 pairs where only CLP is right vs 562 where only Hindi CoT is) and 55.2% with direct answers (+15.1 [+13.6, +16.7]). It beats Hindi CoT on all nine tasks after Holm correction, including the two tasks where the mid report found Hindi CoT worse than direct answering: date subtraction (86.9% vs 66.0%) and interval date (35.3% vs 23.9%). On interval date, however, CLP is still below Hindi direct answers (46.8%).

But it stays well below English. English CoT on the English questions reaches 88.9%, +18.6 [+17.3, +19.9] points above CLP. Counting unanswered responses as wrong changes none of this (CLP 69.9%, Hindi CoT 59.0%; 39 CLP responses have no Answer line, all 39 of them cut off at the token limit).

A.3 The English–Hindi gap across difficulty

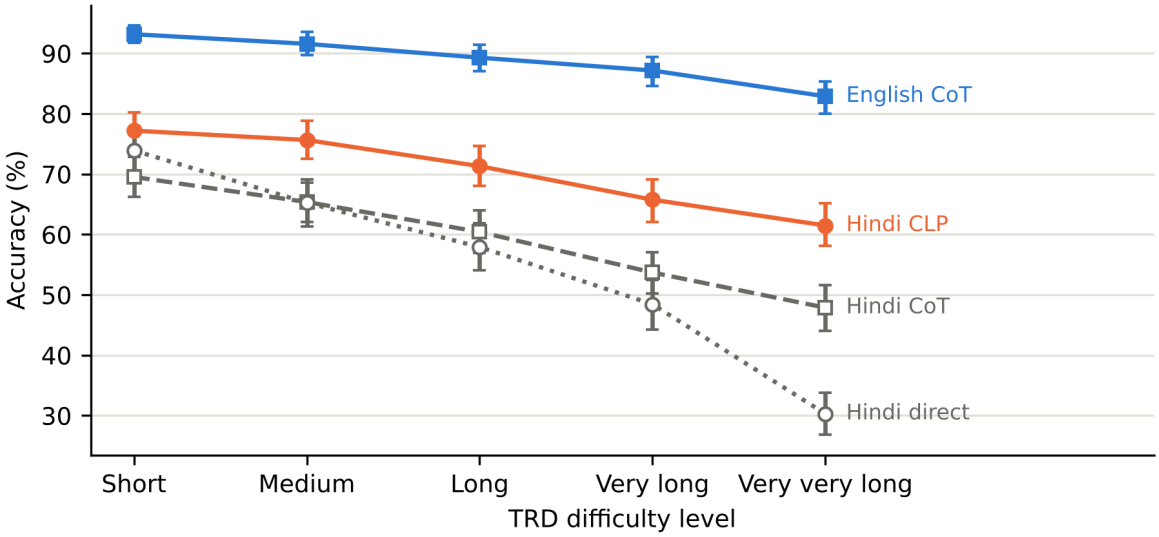


Figure A1. Accuracy by TRD difficulty level, all nine tasks at every level (450 questions per level), with 95% bootstrap CIs. English direct answers are omitted for readability (Table A2 has every value).

Level	English direct	English CoT	Hindi direct	Hindi CoT	Hindi CLP	En CoT − Hi CoT	En CoT − Hi CLP
Short	77.8	93.2	73.9	69.5	77.2	+23.6 [+20.7, +27.0]	+16.0 [+13.0, +19.0]
Medium	69.6	91.6	65.3	65.4	75.7	+26.2 [+23.3, +29.1]	+15.9 [+13.3, +18.7]
Long	61.2	89.3	57.9	60.4	71.3	+28.9 [+25.7, +32.1]	+18.0 [+15.2, +20.7]
Very long	54.3	87.2	48.4	53.7	65.8	+33.5 [+30.1, +36.8]	+21.4 [+18.4, +24.5]
Very very long	35.5	82.9	30.3	47.9	61.5	+35.0 [+31.6, +38.5]	+21.4 [+18.1, +24.6]

Table A2. Accuracy (%) by difficulty level and the gap between English CoT and two Hindi conditions (points, 95% bootstrap CI).

CLP narrows the English–Hindi gap at every level, but the gap still widens with difficulty: English CoT leads CLP by +16.0 points at the short level and by +21.4 at the very-very-long level, against +23.6 and +35.0 for Hindi CoT.

A.4 What the understanding turn does

The understanding turn does more than restate the question. On the three tasks whose answers are dates (date addition, subtraction and recurrence), stage 1 already writes a date other than the ones in the question in 98.6% of responses, and its last such date is the correct answer in 63.1%. Stage 2 then largely follows stage 1 (Table A3).

	Stage 1 ends at the gold date	Stage 1 ends at another date or none	All
Responses	1,414	826	2,240
Stage 2 correct (%)	97.5	20.1	69.0
Stage 2's answer = stage 1's last date (%)	97.6	55.7	82.1

Table A3. CLP on the three date-answer tasks: how stage 2 relates to the last date written in stage 1. Stage 1 often prints "**Answer:**" as a heading with the answer on the next line, so its last date is read from the text rather than from an Answer line.

Times and durations are excluded because this reading is unreliable for them (12-hour clock times, side remarks such as "February has 29 days"); the date reading was correct in all 12 responses checked by hand.

When stage 1 ends at the correct date, stage 2 is right in 97.5% of cases; when it does not, stage 2 recovers in only 20.1%. Of the 695 wrong stage-2 answers, 461 repeat stage 1's last date. Date recurrence shows the consequence most clearly: stage 1 reaches the correct date in only 18.9% of recurrence questions, and CLP ends at 27.0% (Hindi CoT 20.1%, English CoT 90.3%). Over the three date tasks, CLP is correct in 69.0%, against 57.1% for Hindi CoT and 94.6% for English CoT.

Stage 1 is informal: it contains a verification step ("let me verify", "alternatively") in 5.8% of responses, and stage 2 in 26.9%, against 1.3% for Hindi CoT and 86.1% for English CoT. CLP also costs more: a median of 487 completion tokens over the two stages, against 212 for Hindi CoT and 353 for English CoT. Both stages are written almost entirely in English (share of English words 0.977 in stage 1 and 0.999 in stage 2).

These are associations within CLP's own runs, not an experiment on the stages. They are consistent with the following reading: the understanding prompt invites the model to solve the problem informally, and the second turn mostly writes up that first solution instead of solving it again, so a mistake made in stage 1 is rarely corrected. Testing it directly, for example by forbidding an answer in stage 1, is future work.

A.5 Distractors under CLP

Condition	None	Time-related	Unrelated	Drop, time-related	Drop, unrelated
English direct	60.6	59.9	58.6	+0.7 [−1.0, +2.5]	+2.0 [+0.4, +3.7]
English CoT	89.3	88.1	89.2	+1.2 [−0.5, +2.7]	+0.2 [−1.4, +1.6]
Hindi direct	56.0	54.9	54.6	+1.2 [−0.5, +2.9]	+1.5 [−0.3, +3.2]
Hindi CoT	60.2	58.9	59.1	+1.3 [−0.8, +3.6]	+1.2 [−0.9, +3.2]
Hindi CLP	69.3	69.3	72.3	−0.1 [−2.1, +2.0]	−3.1 [−5.1, −1.1]

Table A4. Accuracy (%) without a distractor, with a time-related and with an unrelated distractor, and the drop (accuracy without minus with; positive means the distractor hurt), with 95% bootstrap CIs.

Distractors do not hurt CLP. With the time-related distractor CLP's accuracy is unchanged (drop −0.1 [−2.1, +2.0]), and relative to Hindi direct answers its distraction penalty is −1.2 [−4.1, +1.5] (Hindi CoT: +0.2 [−2.6, +2.9]). With the unrelated distractor CLP is even slightly more accurate than without one (drop −3.1 [−5.1, −1.1]); we have no tested explanation for this. Neither English CoT nor Hindi CoT shows a significant drop with the time-related distractor on this subset either.

Reasoning text	Mentions time-related (placebo)	Mentions unrelated (placebo)	OR, time-related	OR, unrelated
English CoT	17.3 (8.9)	11.0 (6.3)	0.61 [0.39, 0.93]	1.63 [0.90, 2.96]
Hindi CoT	34.0 (10.3)	28.7 (1.9)	0.85 [0.68, 1.07]	1.30 [1.00, 1.70]
CLP stage 1	62.9 (9.4)	83.7 (2.8)	1.02 [0.79, 1.30]	1.65 [1.19, 2.30]
CLP stage 2	17.2 (9.7)	8.7 (2.3)	1.00 [0.72, 1.40]	1.30 [0.81, 2.07]

Table A5. Share (%) of answered responses whose reasoning mentions the distractor (mid-report rule: a content word of the distractor that is not in the question; for CLP the words of the matching English distractor also count). Placebo: the same check

on the same questions without the distractor, i.e. the rate of chance matches such as month names. OR: Mantel–Haenszel odds ratio of a correct answer for responses that mention the distractor versus those that do not, stratified by task and difficulty.

The understanding turn absorbs the distractor. Stage 1 mentions the distractor in most responses (62.9% for the time-related and 83.7% for the unrelated distractor; for example, it translates the sentence or notes that the user forgets how many days a month has), but stage 2 rarely carries it into the solution: 17.2% and 8.7%, against placebo rates of 9.7% and 2.3%. Net of chance matches, that is about 7.5 and 6.5 points, against 23.8 and 26.8 for Hindi CoT traces. In stage 2, mentioning the time-related distractor is not linked to accuracy (OR 1.00 [0.72, 1.40]).

Wrong answers also stop borrowing from the distractor. As in the mid report, wrong Hindi direct answers often contain a number, month or weekday from the time-related distractor (12.6% vs 3.8% placebo, Fisher p <0.001); for wrong CLP answers the rate is 2.6% against 3.1% placebo (p 0.82).

A.6 Does CLP make Hindi fail like English?

Hindi condition (vs English)	Pairs	Both right	Both wrong	Only English right	Only Hindi right	Median φ [IQR]
Hindi direct (vs English direct)	6,743	3,143	2,138	881	581	0.28 [0.14, 0.42]
Hindi CoT (vs English CoT)	6,490	3,778	557	2,000	155	0.07 [-0.01, 0.12]
Hindi CLP (vs English CoT)	6,497	4,412	479	1,366	240	0.10 [-0.03, 0.14]

Table A6. Agreement between each Hindi condition and the English condition on the same question and distractor. φ is the correlation of right/wrong, computed within each task × difficulty cell that has both outcomes in both conditions (43, 35, 35 of 45 cells, in table order) so that differences in difficulty between cells do not inflate it.

CLP brings Hindi accuracy closer to English, but not its failures. Within task and difficulty, whether CLP is right is barely related to whether English CoT is right on the same question (median φ 0.10, like Hindi CoT's 0.07), while direct answers in the two languages fail together more often (0.28). φ cannot be computed where English CoT is right on every question of a cell, which is why 10 of the 45 cells drop out for CLP. CLP's errors also keep a Hindi-like profile: 26.9% of wrong CLP answers are off by one, close to Hindi CoT (21.2%) and far from English CoT (52.3%).

Two task-specific failures from the mid report shrink but remain. On day of week, CLP is right in 31.8% (Hindi CoT 21.0%) and still predicts the day after the correct one in 22.4% of answers with a recognisable weekday (Hindi CoT 25.3%). On date recurrence, 8.1% of CLP answers count the right number of days backwards (Hindi CoT 9.0%, English CoT 0.4%).

A.7 Findings for RQ4

1. **CLP improves Hindi temporal reasoning** by +10.9 points over Hindi CoT and +15.1 over Hindi direct answers, on every task.
2. **It does not close the gap to English:** English CoT is still +18.6 points ahead, and the gap still widens with difficulty.
3. **Stage 2 mostly writes up stage 1's informal solution.** On date tasks, stage 2's answer equals stage 1's last date in most responses and rarely recovers from a wrong one; stage 1 rarely verifies. This plausibly limits what CLP gains, but it was not tested as a cause.

4. **CLP is not more vulnerable to distractors.** The understanding turn deals with the distractor and stage 2 seldom uses it.
5. **CLP makes Hindi accuracy more English-like but not its errors:** failures remain unrelated to English failures and keep a Hindi error profile.

A.8 Limitations

- **Subset.** CLP was run on 2,250 of the 6,750 questions (the first 50 per task and level); all comparisons use the same subset. On it the Hindi CoT distraction penalty is smaller than on the full grid (+0.2 here against +2.2 on all 6,750 questions, from the full-grid analysis), so Part A cannot speak to that effect.
- **One prompt and one model, thinking off.** CLP follows Qin et al.'s wording with our answer-format instructions added to stage 2; other wordings, or Sarvam's own thinking mode, may behave differently. [EDIT: link to the main study's discussion of thinking mode.]
- **Stage-1 measure.** Stage 1 has no reliable Answer line, so its last date is read from the text, only for the three date-answer tasks.
- **Mention rule.** Word matching counts chance matches (the placebo rates) and misses paraphrases.

Part B. RQ5: Do distractor mentions cause reasoning errors?

The mid report found that Hindi CoT traces take up the distractor far more often than English traces and that traces mentioning the time-related distractor have lower odds of a correct answer. That is a correlation: a question that is hard for the model could both draw it into talking about the distractor and lead it to a wrong answer. We ask whether the distractor, and the model's written engagement with it, actually cause these errors, using two intervention experiments on wrong traces with the time-related distractor. [EDIT: merge related work: Shi et al. 2023a; Lanham et al. 2023; Turpin et al. 2023; Yang et al. 2025; Bogdan et al. 2025 (Thought Anchors); Macar et al. 2026 (Thought Branches) [verify].]

B.1 Setup

Traces. From the main runs' CoT responses with the time-related distractor (6,750 English and 6,747 Hindi), we keep wrong answers whose trace mentions the distractor before the sentence that first states the final answer: 526 Hindi and 37 English traces (18.8% and 4.7% of wrong answers). A sentence mentions the distractor if it shares at least two distractor content words absent from the question (month and weekday names excluded), or one word that occurs in fewer than 2% of no-distractor traces. This rule is stricter than the mid-report rule used in Part A.

Model. Both experiments need the model to continue a partial trace, which the Sarvam API cannot do (a pre-filled reply is treated as a finished turn and restarted). We served RedHatAI/sarvam-105b-FP8-dynamic with vLLM 0.31.0 on two H100 GPUs, thinking off as in the main study. On 90 no-distractor calibration questions, the self-hosted model with sampling (temperature 0.8, top-p 0.95) matched the API (71.1% vs 73.3%; English 82.2 vs 84.4; Hindi 60.0 vs 62.2; 89/90 finished). With greedy decoding it fell behind in English (73.3%) and left more answers unfinished (81/90 finished).

B.2 Experiment 1: Deleting the distractor from the trace

Each trace is cut before its answer sentence. **A** keeps the prefix as written, **B** deletes the sentences that mention the distractor, and **C** deletes the same number of length-matched neutral sentences (no digits, no distractor words) as a control for deleting text at all. Requiring neutral sentences leaves 234 traces (222 Hindi, 12 English). The model continued each version once with greedy decoding (G) and once with sampling (S), up to 12,288 tokens; version A was run a second time to measure run-to-run variation.

Run	Language	n	A	B	C	B − C [95% CI]	Only B / only C	p
G (greedy)	Hindi	222	33.3	33.8	33.8	+0.0 [−6.8, +7.2]	32 / 32	1.00
G (greedy)	English	12	25.0	16.7	25.0	−8.3 [−25.0, +0.0]	0 / 1	1.00
S (sampling)	Hindi	222	29.3	32.9	28.4	+4.5 [−2.7, +11.7]	37 / 27	0.26
S (sampling)	English	12	25.0	33.3	25.0	+8.3 [+0.0, +25.0]	1 / 0	1.00

Table B1. Accuracy (%) after continuing each version of the 234 originally wrong traces. p: exact McNemar test, B vs C. English is descriptive only.

Deleting the distractor did not help: in Hindi, B and C recover the same share of traces under greedy decoding (+0.0 points) and an insignificant +4.5 points under sampling (p = 0.26). After deletion the model often wrote the distractor back (35.1% of Hindi B continuations under sampling, about as often as after the unedited prefix, 34.7%; this uses the looser whole-text rule). Single continuations were noisy:

rerunning version A changed right/wrong in 38 of 222 Hindi traces under greedy decoding and 53 under sampling. 102 of 702 continuations (14.5%) under greedy and 40 (5.7%) under sampling were cut off at the token limit, most of them in repetition loops (80 of 102 and 34 of 40).

This null result could mean the mention does not matter, or that the method could not see it: hand-deleting a sentence produces text the model would not have written and that it tends to repair (Macar et al., 2026 [verify]), the distractor remains in the question, one continuation per version is mostly chance, and only 12 English traces survive the neutral-sentence requirement.

B.3 Experiment 2: On-policy resampling

Instead of editing traces we regenerate them from controlled points (Bogdan et al., 2025 [verify]). Let k be the first sentence of a wrong trace that mentions the distractor; each trace yields up to four requests (Table B2), every request is repeated in 7 rounds with a new random seed (sampling as above, up to 4,096 tokens), and accuracy is averaged per trace and arm. All 563 traces are used, because no neutral sentences are needed; in 300 of them (278 Hindi, 22 English) the very first sentence already mentions the distractor, so P and Q start from an empty answer and the whole reasoning is the self-hosted model's own. The design and the primary contrasts (Q − P, M − P and (M − P) − (P − P1), per language) were fixed in the experiment's README before the run; everything else is exploratory.

Arm	Question	Trace prefix	Example: long/date_recurrence_hi_IN/12 (correct of 7)
Q	Without the distractor	Sentences before k	5 / 7
P	With the distractor	Sentences before k	2 / 7
M	With the distractor	Sentences up to and including k	0 / 7
P1	With the distractor	Sentences before k − 1 (only when k ≥ 1)	2 / 7

Table B2. The four arms. Example (Hindi): "The last weekly commitment was in January. Today is 2027-05-16 … what is the date of the 7th occurrence?" (gold 2027-08-15). The original trace's sentence k reads "The last weekly commitment was in January, which means … 31 January 2027 is our start date"; it answered 2027-04-20.

Group	Traces	Q (no distractor)	P (before mention)	M (after mention)	P1 (one sentence earlier)
Hindi, all	526	45.2	40.0	36.7	–
opening mention	278	56.0	50.7	49.1	–
later mention	248	33.1	27.9	22.7	28.6
English, all	37	48.3	31.3	17.0	–
opening mention	22	61.7	37.0	16.2	–
later mention	15	28.6	22.9	18.1	27.6

Table B3. Accuracy (%) of resampled continuations, averaged over 7 runs per trace and arm.

Group	n	Q − P: distractor in question	M − P: written mention	P − P1: ordinary sentence	(M − P) − (P − P1)
Hindi, all	526	+5.3 [+3.2, +7.4], p <0.001	−3.3 [−5.4, −1.2], p 0.002	−0.7 [−3.5, +2.1], p 0.66 (n=248)	−4.6 [−9.6, +0.6], p 0.09 (n=248)

Group	n	Q − P: distractor in question	M − P: written mention	P − P1: ordinary sentence	(M − P) − (P − P1)
opening mention	278	+5.3 [+2.4, +8.2], p <0.001	−1.5 [−4.2, +1.2], p 0.29	–	–
later mention	248	+5.2 [+2.4, +8.1], p <0.001	−5.2 [−8.3, −2.2], p 0.001	−0.7 [−3.5, +2.2], p 0.66	−4.6 [−9.6, +0.5], p 0.09
English, all	37	+17.0 [+8.9, +25.5], p <0.001	−14.3 [−25.1, −4.2], p 0.02	−4.8 [−13.3, +3.8], p 0.43 (n=15)	+0.0 [−22.9, +23.8], p 1.00 (n=15)
opening mention	22	+24.7 [+14.3, +35.1], p <0.001	−20.8 [−33.8, −7.8], p 0.006	–	–
later mention	15	+5.7 [−2.9, +16.2], p 0.36	−4.8 [−22.9, +11.4], p 0.72	−4.8 [−13.3, +2.9], p 0.43	+0.0 [−21.9, +21.9], p 1.00

Table B4. Paired differences (points) with 95% bootstrap CIs (5,000 resamples of traces) and sign-flip permutation p-values (20,000 permutations). The "all" rows are the primary contrasts; contrasts involving P1 use only traces whose mention is not the opening sentence, so for them the "all" and "later mention" rows use the same traces; their intervals differ slightly because each row's bootstrap is drawn separately.

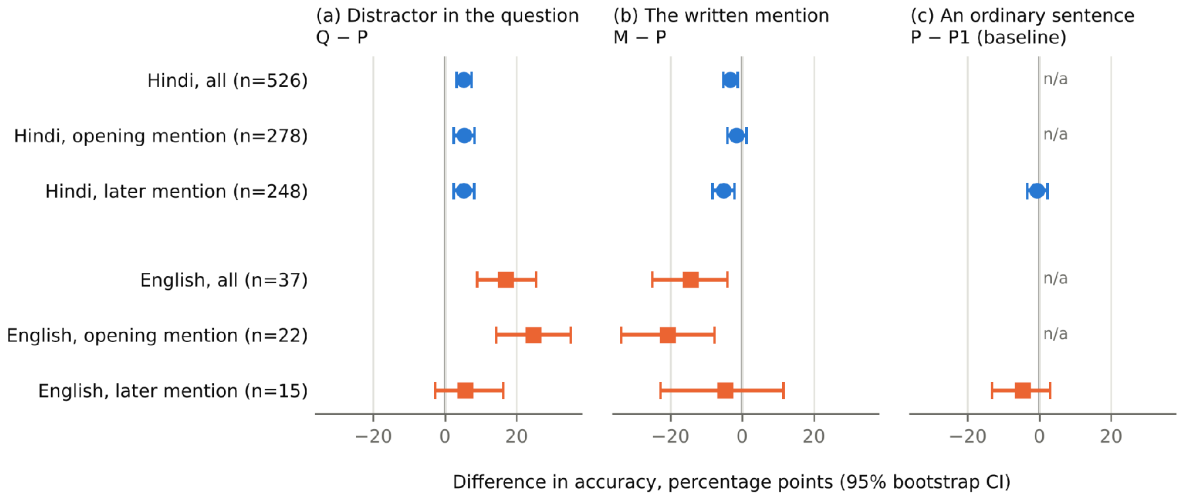


Figure B1. The three comparisons by language and by where the mention occurs (95% CIs). Hindi blue circles, English orange squares.

The distractor causes a minority of these errors. With the reasoning so far held fixed, removing the distractor from the question raises accuracy by +5.3 [+3.2, +7.4] points in Hindi and +17.0 [+8.9, +25.5] in English (both $p < 0.001$). Of the 60.0% of Hindi continuations that are wrong with the distractor, 54.8% stay wrong without it: about one error in eleven is due to the distractor (one in four in English: 68.7% vs 51.7%). Reading a wrong trace that engages with the distractor therefore mostly misattributes the error.

The written mention matters only in the middle of the reasoning. Forcing the model's own mention changes Hindi accuracy by −3.3 [−5.4, −1.2] points ($p = 0.002$). The effect sits in traces whose mention comes later (−5.2 [−8.3, −2.2]), where the model typically builds the distractor into its computation; a mention in the opening sentence, usually a remark about the distractor, has no measurable effect (−1.5 [−4.2, +1.2]). An ordinary sentence at the same place has almost none (−0.7 [−3.5, +2.1]). The primary test that the mention matters more than an ordinary sentence points the same way but is not significant

(−4.6 [−9.6, +0.6], p = 0.09). With six primary tests (Bonferroni threshold 0.0083), Hindi Q − P, Hindi M − P and English Q − P are significant; English M − P (p = 0.02) is not.

Experiment 1's null was real. On the 222 Hindi traces of Experiment 1, forcing the mention makes no difference (M − P −1.0 [−3.9, +1.9]) while removing the distractor from the question does (Q − P +7.6 [+4.6, +10.6]): on those traces the distractor acts through the question, not through the trace's mention of it.

Part of the harm is not finishing. Between 9.4% and 11.9% of continuations per round reached the 4,096-token limit; 554 of 1,205 Hindi and 93 of 232 English cut-off outputs end in an exact repetition loop, while answered continuations rarely approach the limit (99th percentile 3,124 tokens; 0.7% above 3,500). The distractor raises the Hindi rate of unanswered continuations from 5.7% (Q) to 10.0% (P). Counting only answered continuations, Hindi Q − P is +3.6 [+1.5, +5.7] and M − P −3.2 [−5.4, −0.9] (later mentions −5.7 [−9.3, −2.2]; opening −0.9 [−4.0, +1.9]); English Q − P is +13.4 [+3.4, +24.3] and M − P −16.4 [−31.3, −1.7]. All conclusions hold, and about a third of the Hindi distractor effect is non-termination.

Noise and mentions. Across the 7 runs, 372 of 526 Hindi traces were right in some runs and wrong in others under P, so single continuations mostly measure chance. Left free (P), the model mentions the distractor in 57.8% of Hindi continuations (detector false positives on Q: 1.2%).

B.4 Findings for RQ5

1. **A wrong trace that mentions the distractor mostly misattributes the error:** removing the distractor fixes about one in eleven such errors in Hindi and one in four in English.
2. **The distractor acts mainly through the question,** not through the model's mention of it; deleting or forcing the mention usually changes nothing, and the model re-mentions it after deletion.
3. **Mentions matter when the distractor becomes part of the computation** (later mentions, about 5 points in Hindi); the direct test against an ordinary sentence is a non-significant trend.
4. **Part of the harm is not finishing:** the distractor nearly doubles Hindi non-termination.
5. **Hindi engages the distractor more often, English is hurt more when it does** (37 English traces: suggestive).

B.5 Limitations

- Only originally wrong traces; resampled accuracy is far above zero partly by regression to the mean (arm comparisons are unaffected).
- Continuations come from an FP8 deployment matched to the API by calibration; the source traces came from the API; thinking was off.
- For later mentions the prefix was written under the distractor question and may already carry its influence, so Q − P may understate it.
- Lexical mention detector; 4,096-token limit with unfinished continuations counted as wrong (the answered-only analysis gives the same conclusions); 37 English traces; only the time-related distractor.

Appendix A. Claims and evidence (for the presentation)

Claim	Evidence	Strength
CLP improves Hindi accuracy over Hindi CoT and direct answers.	Table A1: +10.9 [+9.6, +12.2] vs CoT, +15.1 [+13.6, +16.7] vs direct; better on all 9 tasks.	Strong
CLP narrows but does not close the English gap, which still grows with difficulty.	Table A2: gap +16.0 (short) to +21.4 (very very long).	Strong
Stage 2 mostly follows stage 1's informal answer.	Table A3: 97.5% vs 20.1% correct; 461 of 695 wrong answers repeat stage 1.	Strong as association; cause untested
Distractors do not hurt CLP.	Time-related drop −0.1 [−2.1, +2.0]; penalty vs direct −1.2 [−4.1, +1.5].	Moderate (CIs about ±2–3 points)
CLP's errors stay Hindi-like.	φ 0.10 vs English; off-by-one share 26.9% vs 52.3% for English CoT.	Moderate
The distractor causes a minority of the errors where wrong traces mention it.	Q − P: Hindi +5.3 [+3.2, +7.4], English +17.0 [+8.9, +25.5].	Strong (pre-registered)
Forcing the model's mention lowers Hindi accuracy, only for later mentions.	M − P −3.3 [−5.4, −1.2]; later −5.2 [−8.3, −2.2]; opening −1.5 [−4.2, +1.2].	Strong overall; split exploratory
The mention matters more than an ordinary sentence.	−4.6 [−9.6, +0.6], p = 0.09.	Not significant
Deleting the distractor from the trace does not help (Experiment 1).	Hindi B − C +0.0 (G), +4.5 (S).	Null; confirmed on the same traces by Experiment 2

References to add [verify]

- Libo Qin, Qiguang Chen, Fuxuan Wei, Shijue Huang, Wanxiang Che. 2023. Cross-lingual Prompting: Improving Zero-shot Chain-of-Thought Reasoning across Languages. EMNLP 2023. (already in the mid-report bibliography)
- Paul C. Bogdan, Uzay Macar, Neel Nanda, Arthur Conmy. 2025. Thought Anchors: Which LLM Reasoning Steps Matter? arXiv:2506.19143.
- Uzay Macar, Paul C. Bogdan, et al. 2026. Thought Branches: Interpreting LLM Reasoning Requires Resampling. ICLR 2026. arXiv:2510.27484.
- Tamera Lanham et al. 2023. Measuring Faithfulness in Chain-of-Thought Reasoning. arXiv:2307.13702.
- Sohee Yang et al. 2025. How Well Can Reasoning Models Identify and Recover from Unhelpful Thoughts? arXiv:2506.10979.

Part C. Does reasoning in English (en-CoT) help Hindi temporal reasoning?

C.1 Method

In en-CoT (Shi et al., 2023) the question stays in Hindi, but the model is asked to reason step by step in English before giving its final answer line. Unlike CLP, it is a single turn. We used the main study's settings: Sarvam-105B through the API, thinking off, temperature 0, up to 4,096 tokens. Day-of-week answers are still asked for as Hindi weekday names and scored by weekday.

en-CoT was run on questions 0–49 of every task and difficulty level, the same 2,250 questions as CLP. Each question was asked with no distractor, a time-related distractor and an unrelated distractor, giving 6,750 responses. We compare it with English and Hindi direct answers, English and Hindi CoT, and CLP, all on exactly these questions.

The pipeline and statistics are the same as in Part A: the full-grid answer extraction and scoring, exact McNemar tests on paired responses with Holm correction over the nine tasks, and 95% confidence intervals from a 2,000-sample bootstrap over questions. The model followed the instruction: 99.9% of the words in en-CoT reasoning are English.

C.2 Accuracy

Task	English direct	English CoT	Hindi direct	Hindi CoT	Hindi CLP	Hindi en-CoT	en-CoT − Hindi CoT	en-CoT − Hindi direct	en-CoT − CLP
Date addition	85.6	98.0	82.8	85.1	92.7	95.7	+10.7*	+12.9*	+3.1*
Date subtraction	86.0	95.2	84.8	66.0	86.9	94.0	+28.0*	+9.2*	+7.1*
Date duration	84.0	99.3	78.9	88.3	92.7	97.7	+9.5*	+18.8*	+5.1*
Date recurrence	22.1	90.3	2.8	20.1	27.0	70.3	+50.2*	+67.5*	+43.4*
Day of week	14.9	69.5	15.2	21.0	31.8	52.5	+31.5*	+37.3*	+20.7*
Interval date	43.2	50.7	46.8	23.9	35.3	44.1	+20.3*	−2.7	+8.8*
Time addition	70.5	95.9	75.6	73.5	88.2	92.6	+19.2*	+17.0*	+4.4*
Time subtraction	64.7	97.7	64.9	75.2	90.1	95.7	+20.5*	+30.8*	+5.6*
Time duration	65.7	98.9	44.7	80.1	86.7	98.1	+18.1*	+53.5*	+11.5*
All tasks	**59.7**	**88.9**	**55.2**	**59.4**	**70.3**	**82.9**	**+23.5***	**+27.7***	**+12.6***

*Table C1. Accuracy (%) of answered responses on 2,250 questions × 3 distractor conditions, pooled over difficulty and distractor. Differences are in points. * marks p < 0.05 in an exact McNemar test on paired responses, Holm-corrected over the nine tasks; the All tasks row is uncorrected.*

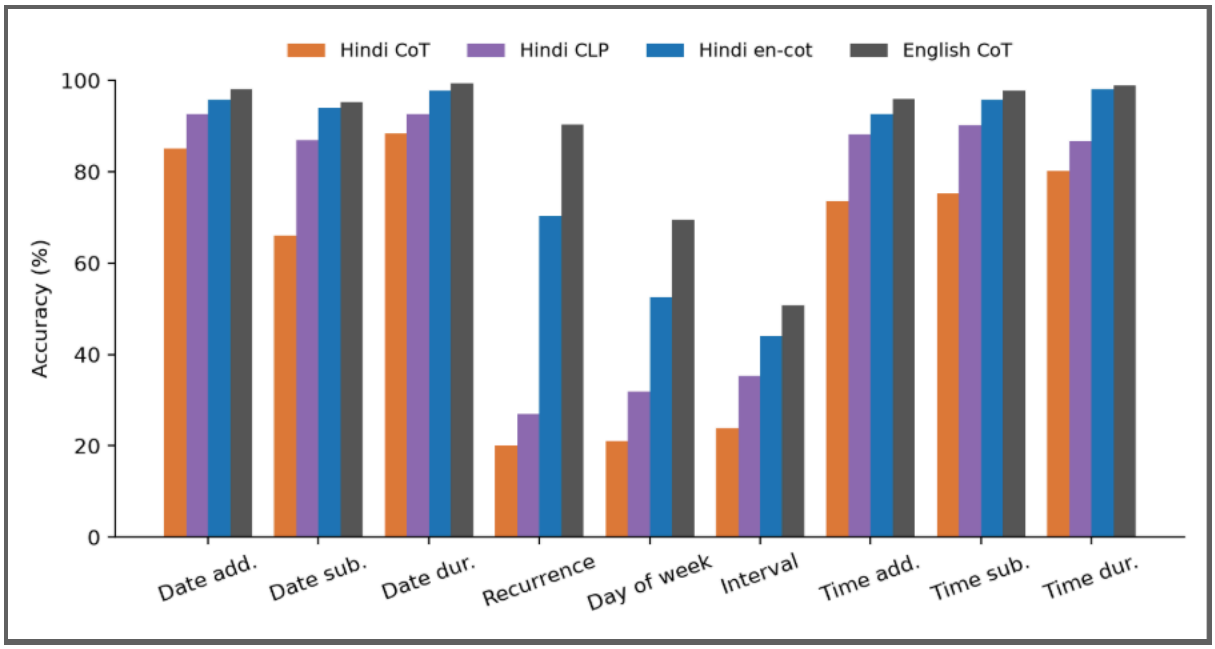


Figure C2. Accuracy by task for Hindi CoT, CLP, en-CoT and English CoT, on the same 2,250 questions.

en-CoT helps Hindi a lot, and more than CLP. It answers 82.9% of the Hindi questions correctly:

- against Hindi CoT (59.4%): +23.5 [+22.1, +24.9] points, with 1,742 pairs where only en-CoT is right and 272 where only Hindi CoT is;
- against Hindi direct answers (55.2%): +27.7 [+25.8, +29.5];
- against CLP (70.3%): +12.6 [+11.2, +13.8], with 1,158 pairs against 400.

It beats both Hindi CoT and CLP on all nine tasks after Holm correction. The two tasks where the mid report found Hindi CoT worse than direct answering both recover. Date subtraction reaches 94.0% (Hindi CoT 66.0%). Interval date reaches 44.1% (Hindi CoT 23.9%), level with Hindi direct answers (46.8%; the difference is not significant). The biggest jumps are on recurrence (20.1% → 70.3%) and day of week (21.0% → 52.5%).

It comes close to English. English CoT is still ahead, but only by +6.0 [+5.0, +7.0] points. Counting unanswered responses as wrong does not change the picture: en-CoT 80.8%, CLP 69.9%, Hindi CoT 59.0%. 174 en-CoT responses have no Answer line, all of them cut off at the token limit, against 43 for Hindi CoT and 221 for English CoT.

C.3 The English–Hindi gap across difficulty

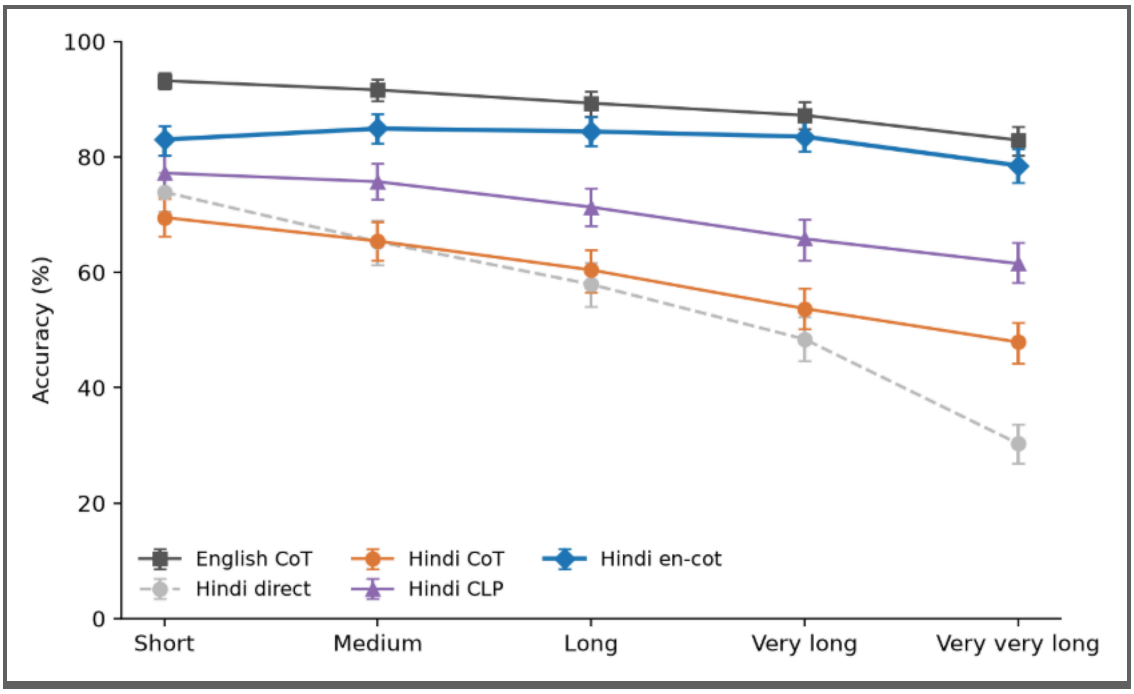


Figure C1. Accuracy by difficulty level, all nine tasks at every level (450 questions per level), with 95% bootstrap CIs.

Level	English CoT	Hindi CoT	Hindi CLP	Hindi en-CoT	En CoT − Hi CoT	En CoT − Hi en-CoT
Short	93.2	69.5	77.2	83.0	+23.6 [+20.6, +26.7]	+10.2 [+7.8, +12.7]
Medium	91.6	65.4	75.7	84.9	+26.2 [+23.3, +29.0]	+6.7 [+4.6, +8.7]
Long	89.3	60.4	71.3	84.4	+28.9 [+25.7, +32.2]	+4.9 [+3.1, +6.8]
Very long	87.2	53.7	65.8	83.5	+33.5 [+30.1, +37.0]	+3.6 [+1.5, +6.1]
Very very long	82.9	47.9	61.5	78.5	+35.0 [+31.5, +38.4]	+4.3 [+1.9, +6.6]

Table C2. Accuracy (%) by difficulty level, and the gap between English CoT and two Hindi conditions (points, 95% bootstrap CI).

With en-CoT the gap to English shrinks as questions get harder, instead of growing. It goes from +10.2 points at the short level to +4.3 at the very-very-long level. For Hindi CoT it grows from +23.6 to +35.0, and for CLP from +16.0 to +21.4 (Part A). en-CoT stays between 78.5% and 84.9% at every level. Its weakest level against English is short, mostly because of the recurrence questions that ask for the 1st or 2nd occurrence (C.6).

C.4 Distractors under en-CoT

Condition	None	Time-related	Unrelated	Drop, time-related	Drop, unrelated
English direct	60.6	59.9	58.6	+0.7 [−1.0, +2.6]	+2.0 [+0.3, +3.8]
English CoT	89.3	88.1	89.2	+1.2 [−0.4, +2.7]	+0.2 [−1.4, +1.7]
Hindi direct	56.0	54.9	54.6	+1.2 [−0.5, +2.9]	+1.5 [−0.3, +3.2]
Hindi CoT	60.2	58.9	59.1	+1.3 [−0.9, +3.5]	+1.2 [−0.9, +3.3]

Hindi CLP	69.3	69.3	72.3	−0.1 [−2.2, +2.0]	−3.1 [−5.1, −1.2]
Hindi en-CoT	84.1	81.0	83.6	+3.2 [+1.3, +4.9]	+0.6 [−1.2, +2.3]

Table C3. Accuracy (%) without a distractor, with a time-related distractor and with an unrelated one, and the drop (accuracy without minus with; positive means the distractor hurt), with 95% bootstrap CIs.

en-CoT is the only condition on this subset that loses accuracy with the time-related distractor: 84.1% → 81.0%, a drop of +3.2 [+1.3, +4.9] points. It is small, and the unrelated distractor makes no difference. We have not tested why. One possibility is that the Hindi distractor gets carried into the English reasoning.

Reasoning text	Mentions time-related (placebo)	Mentions unrelated (placebo)	OR, time-related	OR, unrelated
English CoT	17.3 (8.9)	11.0 (6.3)	0.61 [0.39, 0.93]	1.63 [0.90, 2.96]
Hindi CoT	34.0 (10.3)	28.7 (1.9)	0.85 [0.68, 1.07]	1.30 [1.00, 1.70]
Hindi en-CoT	22.0 (9.3)	17.0 (5.5)	0.82 [0.59, 1.13]	0.91 [0.58, 1.43]

Table C4. Share (%) of answered responses whose reasoning mentions the distractor, using the mid-report rule; for en-CoT the words of the matching English distractor also count, as for CLP. Placebo: the same check on the same questions without the distractor. OR: Mantel–Haenszel odds ratio of a correct answer for responses that mention the distractor against those that do not, stratified by task and difficulty.

en-CoT talks about the distractor about half as often as Hindi CoT. After subtracting chance matches, en-CoT mentions it 12.7 points above placebo (time-related) and 11.5 points (unrelated), against 23.7 and 26.8 for Hindi CoT. Mentioning it is not linked to getting the answer wrong (OR 0.82 [0.59, 1.13] and 0.91 [0.58, 1.43]).

C.5 Does en-CoT make Hindi fail like English?

Hindi condition (vs English)	Pairs	Both right	Both wrong	Only English right	Only Hindi right	Median φ [IQR]
Hindi direct (vs English direct)	6,743	3,143	2,138	881	581	0.28 [0.14, 0.42]
Hindi CoT (vs English CoT)	6,490	3,778	557	2,000	155	0.07 [−0.01, 0.12]
Hindi en-CoT (vs English CoT)	6,395	5,033	370	672	320	0.02 [−0.03, 0.19]

Table C5. Agreement between each Hindi condition and the English condition on the same question and distractor. φ is computed within each task × difficulty cell where both conditions have both outcomes (43, 35 and 34 of 45 cells, in table order).

en-CoT gets close to English accuracy, but it still fails on different questions. Whether en-CoT is right is barely related to whether English CoT is right on the same question (median φ 0.02, like Hindi CoT's 0.07).

Its errors look more like English ones, though:

- **Off by one:** 42.3% of wrong en-CoT answers are off by one. That is between Hindi CoT (21.2%) and English CoT (52.3%), and well above CLP (26.9%).

- **Checking its work:** en-CoT includes a verification step ("let me verify", "double-check") in 74.2% of responses, against 1.3% for Hindi CoT and 86.1% for English CoT.
- **Length:** a median of 311 completion tokens, against 212 for Hindi CoT, 353 for English CoT and 487 for CLP's two stages.

C.6 Two failures that stay

Day of week: one day late. en-CoT is right on 52.5% of day-of-week questions (Hindi CoT 21.0%, English CoT 69.5%). It is still one day late in 24.0% of answers that name a weekday (Hindi CoT 25.3%, English CoT 16.2%). It is the same slip English CoT makes: when counting the days from 1 January, the model counts 1 January itself.

Recurrence: going backwards. 8.3% of en-CoT recurrence answers count the right number of days in the wrong direction, almost the same as Hindi CoT (9.0%) and far above English CoT (0.4%). en-CoT reasons in English, so this comes from reading the Hindi question, not from reasoning in Hindi. The model reads "आज को छोड़कर, 1वीं आवृत्ति" (excluding today, the 1st occurrence) as the occurrence before today. For example: "Given the phrasing 'excluding today,' this suggests we need to find what comes before." This matches the full-grid finding that these errors sit in the k = 1 and k = 2 template cells. [EDIT: cross-reference]

C.7 Findings

- en-CoT improves Hindi temporal reasoning by +23.5 points over Hindi CoT, +27.7 over Hindi direct answers and +12.6 over CLP, on every task.
- It comes within 6.0 points of English CoT, and unlike CLP the gap shrinks as questions get harder (+10.2 at short, +4.3 at very very long).
- What is left comes mostly from reading the Hindi question (recurrences counted backwards) and from a counting slip English CoT makes too (one day late on day of week).
- en-CoT's errors and habits (off-by-one errors, checking its work) look more like English CoT's, but it still fails on different questions from English CoT.
- The time-related distractor costs en-CoT a small but significant 3.2 points. Mentioning the distractor is not linked to wrong answers.
- One plain English reasoning turn beats CLP's two turns on these questions. One possible reason, which we have not tested: CLP's first turn solves the problem informally and the second turn mostly copies it (Part A.4), while en-CoT reasons once, step by step, and usually checks its work.

C.8 Limitations

- **Subset:** en-CoT was run on 2,250 of the 6,750 questions (the first 50 per task and level). All comparisons use this subset.
- **One prompt, one model, thinking off:** other wordings, or Sarvam's thinking mode, may behave differently.
- **Unanswered responses:** 174 en-CoT responses were cut off at 4,096 tokens and count as unanswered in Table C1; strict accuracy is 80.8%.
- **Mention rule:** word matching counts chance matches (the placebo rates) and misses paraphrases. For en-CoT we count both the Hindi and English distractor words.

- **Distractor effect:** we did not test why the time-related distractor hurts en-CoT.

Appendix C. Claims and evidence (for the presentation)

Claim	Evidence	Strength
en-CoT improves Hindi accuracy over Hindi CoT, Hindi direct and CLP	Table C1: +23.5, +27.7 and +12.6 points; better than Hindi CoT and CLP on all 9 tasks	Strong
en-CoT nearly closes the gap to English, and the gap shrinks with difficulty	Table C2: +10.2 (short) to +4.3 (very very long)	Strong
What's left comes from reading Hindi, plus a shared counting slip	Backwards recurrences 8.3% vs 0.4% English; one day late 24.0% vs 16.2%	Moderate
en-CoT's errors and habits look more like English	Off-by-one 42.3% of errors; verification 74.2%	Moderate
en-CoT still fails on different questions from English	φ 0.02 vs English CoT	Moderate
The time-related distractor slightly hurts en-CoT	Drop +3.2 [+1.3, +4.9]	Moderate (small effect, cause untested)